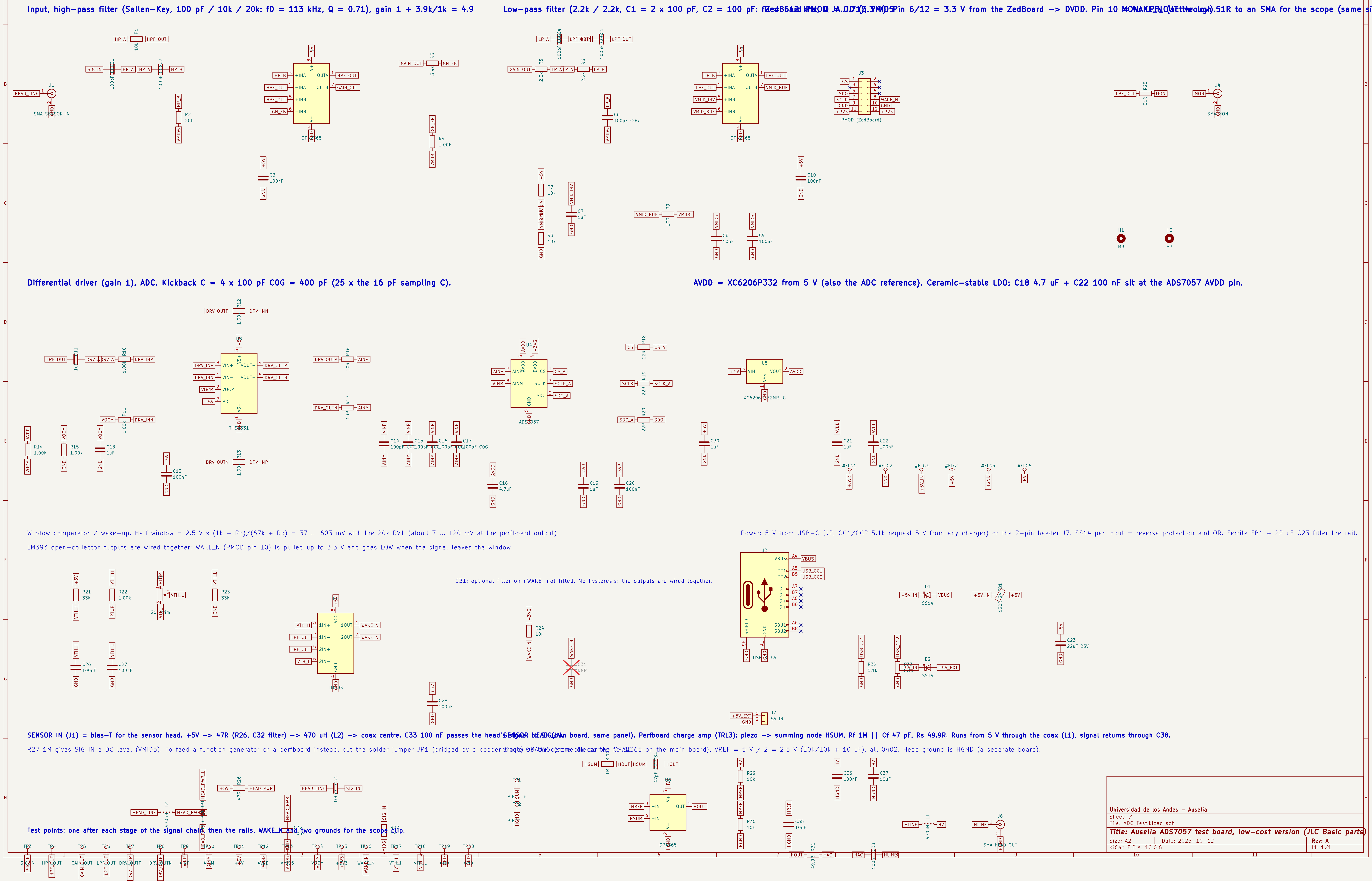
ADS7057 test board (low cost)

Sensor head (SMA, bias–T) –> 2nd–order HPF 113 kHz –> gain x4.9 –> 2nd–order LPF 512 kHz –> THS4531 –> 10R/400pF –> ADS7057 –> ZedBoard PMOD.

AFE v0.1 topology rebuilt around JLC Basic parts (100 pF C0G, E12 resistors, AMS1117, LM393). Power: 5 V (USB–C or header) –> SS14 –> ferrite –> +5V, AVDD = XC6206 3.3 V from 5 V. DVDD = PMOD 3.3 V.



Input, high–pass filter (Sallen–Key, 100 pF / 10k / 20k: f0 = 113 kHz, Q = 0.71), gain 1 + 3.9k/1k = 4.9

Low–pass filter (2.2k / 2.2k, C1 = 2 x 100 pF, C2 = 100 pF: f0 = 512 kHz, Q = 0.71), VMID5

ZedBoard PMOD JA.JD7 (3.3 V). Pin 6/12 = 3.3 V from the ZedBoard –> DVDD. Pin 10 = WAKE_N (active low).

MONITOR: LPF_OUT through 51R to an SMA for the scope (same signal

Differential driver (gain 1), ADC. Kickback C = 4 x 100 pF C0G = 400 pF (25 x the 16 pF sampling C).

AVDD = XC6206P332 from 5 V (also the ADC reference). Ceramic–stable LDO; C18 4.7 uF + C22 100 nF sit at the ADS7057 AVDD pin.

Window comparator / wake–up. Half window = 2.5 V x (1k + Rp)/(67k + Rp) = 37 ... 603 mV with the 20k RV1 (about 7 ... 120 mV at the perfboard output).

LM393 open–collector outputs are wired together: WAKE_N (PMOD pin 10) is pulled up to 3.3 V and goes LOW when the signal leaves the window.

C31: optional filter on nWAKE, not fitted. No hysteresis: the outputs are wired together.

Power: 5 V from USB–C (J2, CC1/CC2 5.1k request 5 V from any charger) or the 2–pin header J7. SS14 per input = reverse protection and OR. Ferrite FB1 + 22 uF C23 filter the rail.

SENSOR IN (J1) = bias–T for the sensor head. +5V –> 47R (R26, C32 filter) –> 470 uH (L2) –> coax centre. C33 100 nF passes the head's signal to SIG_IN.

R27 1M gives SIG_IN a DC level (VMID5). To feed a function generator or a perfboard instead, cut the solder jumper JP1 (bridged by a copper trace)

SENSOR HEAD (own board, same panel). Perfboard charge amp (TRL3): piezo –> summing node HSUM, Rf 1M || Cf 47 pF, Rs 49.9R. Runs from 5 V through the coax (L1), signal returns through C38.

Single OPA365 (same die as the OPA2365 on the main board), VREF = 5 V / 2 = 2.5 V (10k/10k + 10 uF), all 0402. Head ground is HGND (a separate board).

Test points: one after each stage of the signal chain, then the rails, WAKE_N and two grounds for the scope clip.

Universidad de los Andes – Auselia
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File: ADC_Test.kicad_sch
Title: Auselia ADS7057 test board, low–cost version (JLC Basic parts)
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